

APM1110 - FA6

Betita, Jazel

2026-09-20

I. Geometric Distribution

Provide an R code for the geometric distribution. The geometric distribution is a probability distribution that models the number of trials required to achieve the first success in a sequence of Bernoulli trials, where each trial has a constant probability of success.

1. Set the probability to $p <- 0.2$

```
p <- 0.2
```

2. Generate 1000 random variables from the geometric distribution.

```
x <- rgeom(1000, prob = p) + 1
```

3. Calculate basic statistics:

```
mean_x <- mean(x)
var_x <- var(x)
sd_x <- sd(x)
```

4. Print the results in item 3 with the following output format:

```
cat("Number of trials required to achieve first success:", x[1], "\n")
```

```
## Number of trials required to achieve first success: 5
```

```
cat("Mean:", round(mean_x, 2), "\n")
```

```
## Mean: 5.15
```

```
cat("Variance:", round(var_x, 2), "\n")
```

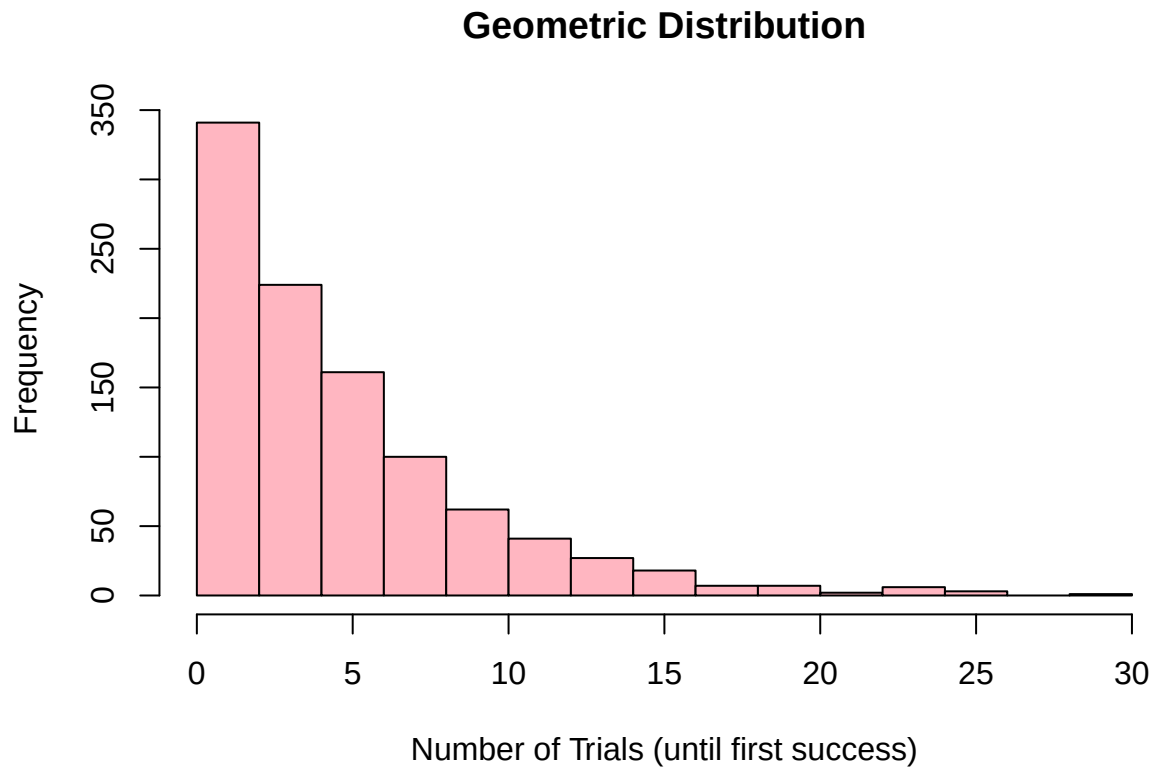
```
## Variance: 19.51
```

```
cat("Standard deviation:", round(sd_x, 2), "\n")
```

```
## Standard deviation: 4.42
```

5. Plot the histogram of the results.

```
hist(x,
     main = "Geometric Distribution",
     xlab = "Number of Trials (until first success)",
     ylab = "Frequency",
     col = "lightpink")
```



```
cat("Number of trials required to achieve first success:", x[1], "\n")
```

```
## Number of trials required to achieve first success: 5
```

II. Hypergeometric Distribution

Consider a plant manufacturing IC chips which 10% are expected to be defective. The chips are packed in boxes for export. Before transportation, a sample is drawn from each box. Estimate the probability that the sample contains more than 10% defectives, when:

1. A sample of 10 is selected from a box of 40;

```
samp <- 10

total1 <- 40
defec1 <- 4

num1 <- 1 - phyper(1, defec1, total1 - defec1, samp)
```

```
cat("Probability that sample contains more than 10% defectives (from a box of 40):",  
    round(num1, 2), "\n")
```

```
## Probability that sample contains more than 10% defectives (from a box of 40): 0.26
```

The possibility of the sample that includes more than 10% defected chips is 0.26. Which means, there is a 26% chance that more than 1 chip is defective among the 10 chips selected.

2. A sample of 10 is selected from a box of 5000.

```
total2 <- 5000  
defec2 <- 500  
  
num2 <- 1 - phyper(1, defec2, total2 - defec2, samp)  
  
cat("Probability that sample contains more than 10% defectives (from a box of 5000):",  
    round(num2, 2), "\n")
```

```
## Probability that sample contains more than 10% defectives (from a box of 5000): 0.26
```

The probability is 0.26. There is a 26% chance that more than 1 chip is defective among the 10 chips selected.

The probabilities of the two boxes are similar because both contains 10% defective rate.